# NIRT SOLAR EPC — MASTER PIPELINE & PROCESS STEPS

**OPERATIONAL ACTIONS (INPUTS & UPLOADS), GATES, AND AUDITED EDIT PROTOCOL**

**1. Lead → 2. Qualify (A-C/D) → 3. Survey → 4. Design → 5. Quote → 6. Order → 7. 80% Advance → 8. DISCOM → 9. Procure → 10. Delivery → 11. Install → 12. 10% On Installation → 13. Net Meter → 14. 10% Final → 15. Subsidy DBT**

| Pipeline / Phase | Step | Process Step | Responsible Role | Action: Inputs & Uploads | Mandatory Gate & Edit Rule |
|---|---|---|---|---|---|
| **1. Sales & Feasibility** | **01** | **Lead Inward & Intake** | Sales Executive | **Input:** Name, Phone, Address, Circle, EB #<br>**Upload:** None (Optional Lead card) | **Gate:** Deduplicated Lead Created<br>**Edit:** Edit with Reason (Logged to Audit) |
| **1. Sales & Feasibility** | **02** | **Lead Qualification (Viability)** | Sales Executive | **Input:** Load, Units, Roof Type, Status<br>**Outcome:** Qualified (A/B/C) / Unqualified (D) | **Gate:** Viability Profile / Drop Reason<br>**Edit:** Edit with Reason (Logged to Audit) |
| **1. Sales & Feasibility** | **03** | **Site Survey & Feasibility Audit** | Site Surveyor | **Input:** Tape dims, Shadow, Cable mtrs<br>**Upload:** 4 Photos (Roof, EB, DB, Wall) | **Gate:** Survey Dossier Approved<br>**Edit:** Edit with Reason (Logged to Audit) |
| **1. Sales & Feasibility** | **04** | **System Design & Engineering DPR** | Design Engineer | **Input:** kW sizing, Inverter model, Strings<br>**Upload:** Single Line Diagram (SLD), CAD | **Gate:** Verified SLD & System DPR<br>**Edit:** Edit with Reason (Logged to Audit) |
| **1. Sales & Feasibility** | **05** | **Commercial Quotation Sizing** | Sales Executive | **Input:** BOM pricing, Discount, Terms<br>**Upload:** None (Auto-generates PDF) | **Gate:** Formal Quotation Dispatched<br>**Edit:** Edit with Reason (Logged to Audit) |
| **1. Sales & Feasibility** | **06** | **Customer Order Confirmation** | Sales / Customer | **Input:** Payment mode, Target date<br>**Upload:** Signed Work Order copy | **Gate:** Project ID Generated (NIRT-XXXX)<br>**Edit:** Edit with Reason (Logged to Audit) |
| **2. Advance & Approvals** | **07** | **80% Advance Receipt Gatekeeper** | Accounts Executive | **Input:** Bank UTR, Amount, Date, Mode<br>**Upload:** Bank credit counterfoil / slip | **Gate:** Stage 7 Cleared (Hard Lock Off)<br>**Edit:** Edit with Reason (Logged to Audit) |
| **2. Advance & Approvals** | **08** | **Document Locker KYC Assembly** | Liaison Officer | **Input:** Aadhaar #, Consumer #, PAN<br>**Upload:** KYC, EB Bill, Cheque, Tax Receipt | **Gate:** Complete Document Dossier<br>**Edit:** Edit with Reason (Logged to Audit) |
| **2. Advance & Approvals** | **09** | **TANGEDCO DISCOM Portal Filing** | Liaison Officer | **Input:** Portal App ID, Submission date<br>**Upload:** DISCOM acknowledgment receipt | **Gate:** Online Application Verified<br>**Edit:** Edit with Reason (Logged to Audit) |
| **2. Advance & Approvals** | **10** | **Technical Feasibility Sanction** | Liaison / DISCOM | **Input:** Sanction Ref #, Sanctioned kW<br>**Upload:** TANGEDCO Feasibility Sanction | **Gate:** Grid Feasibility Sanctioned<br>**Edit:** Edit with Reason (Logged to Audit) |
| **3. Procurement & Store** | **11** | **Vendor PO Issuance** | Procurement Exec | **Input:** PO #, Vendor name, Module/Inv specs<br>**Upload:** Vendor Purchase Order PDF | **Gate:** Factory PO Dispatched<br>**Edit:** Edit with Reason (Logged to Audit) |
| **3. Procurement & Store** | **12** | **Warehouse QC & Inwarding (GRN)** | Store Executive | **Input:** GRN #, Serial numbers, QC status<br>**Upload:** QC Pass Sheet & Delivery Challan | **Gate:** Inward GRN Tagged to Project<br>**Edit:** Edit with Reason (Logged to Audit) |
| **3. Procurement & Store** | **13** | **Material Dispatch to Site** | Store / Logistics | **Input:** Vehicle #, Driver phone, Dispatch date<br>**Upload:** Site Delivery Challan / E-way Bill | **Gate:** Materials Dispatched to Site<br>**Edit:** Edit with Reason (Logged to Audit) |
| **4. EPC Site Execution** | **14** | **Civil Structure & Module Mounting** | EPC Site Engineer | **Input:** Torque check Nm, Tilt angle deg<br>**Upload:** 4 Erection Milestone Photos | **Gate:** Structural Sign-off Recorded<br>**Edit:** Edit with Reason (Logged to Audit) |
| **4. EPC Site Execution** | **15** | **Electrical Cabling & Inverter Setup** | EPC Site Engineer | **Input:** Earthing resistance Ohms (<5), Polarity<br>**Upload:** Inverter display photo, Earth report | **Gate:** Electrical Readiness Signed<br>**Edit:** Edit with Reason (Logged to Audit) |
| **4. EPC Site Execution** | **16** | **10% On Installation Receipt** | Accounts Executive | **Input:** Installation UTR, Amount, Receipt #<br>**Upload:** Bank credit advice / slip | **Gate:** 90% Cumulative Funds Received<br>**Edit:** Edit with Reason (Logged to Audit) |
| **4. EPC Site Execution** | **17** | **TANGEDCO Net Metering & Sync** | Liaison / DISCOM | **Input:** Bi-directional Meter #, Sync date<br>**Upload:** DISCOM Commissioning Report | **Gate:** Grid Synchronized & Generating<br>**Edit:** Edit with Reason (Logged to Audit) |
| **5. Closure & Subsidy** | **18** | **10% Final Balance Collection** | Accounts Executive | **Input:** Final UTR, Receipt #, Balance Zero<br>**Upload:** Final Tax Invoice & Zero Balance | **Gate:** Commercial Closure (Balance = Rs. 0)<br>**Edit:** Edit with Reason (Logged to Audit) |
| **5. Closure & Subsidy** | **19** | **PM Surya Ghar Subsidy Submission** | Subsidy Desk | **Input:** National Portal Claim Ref, Bank IFSC<br>**Upload:** Combined ICR + Meter photo | **Gate:** Portal Subsidy Claim Lodged<br>**Edit:** Edit with Reason (Logged to Audit) |
| **5. Closure & Subsidy** | **20** | **DBT Subsidy Credit & Formal Closure** | Central PMO | **Input:** Subsidy DBT UTR, Sanctioned amount<br>**Upload:** Bank DBT credit proof / statement | **Gate:** Triple Closure (Tech+Comm+Subsidy)<br>**Edit:** Edit with Reason (Logged to Audit) |

**MANDATORY EDIT & AUDIT POLICY: ZERO DELETE OPTION • EDITS PERMITTED AT ALL STEPS WITH MANDATORY REASON ENTRY • ALL CORRECTIONS, INPUTS & UPLOADS RECORDED FORENSICALLY IN PMO AUDIT LOG**

# NIRT SOLAR EPC — SIMPLE OPERATIONS GUIDE

## SECTOR EXECUTION & THE 5 CORE OPERATIONAL PIPELINES

**Core Operating Reality:** Solar projects move across **5 synchronized pipelines** covering **20 sequential steps**. Every department handles specific steps, ensuring complete visibility from lead intake to final central subsidy credit.

### THE 4 KEY SECTORS NIRT EXECUTES:

- **Residential Homes:** Concrete slab & sloped roofs, wiping out domestic power bills under PM Surya Ghar.
- **Commercial Complexes:** Hospitals, schools, colleges, showrooms, and retail hubs cutting daytime commercial tariffs.
- **Industrial Factories & Mills:** Manufacturing sheds, spinning mills, and cold storages slashing peak demand charges.
- **Farm Lands & Agro-Solar:** Agricultural borewells, solar water pumps (PM-KUSUM), and ground installations.

### THE 5 OPERATIONAL PIPELINES (HOW WORK MOVES ACROSS STEPS 01 TO 20):

**1. Sales & Feasibility Pipeline (Steps 01 to 06):** Lead intake → Qualification (A/B/C/D) → Site survey & 4 photos → System SLD & DPR → Quotation sizing → Customer Order.

**2. Advance & Approvals Pipeline (Steps 07 to 10):** Accounts clears 80% Advance UTR → KYC Locker → TANGEDCO online portal filing → Grid feasibility sanction.

**3. Procurement & Store Pipeline (Steps 11 to 13):** Tier-1 vendor PO issuance → Warehouse QC & serial numbering (GRN) → Dispatch to site with delivery challan.

**4. EPC Site Execution Pipeline (Steps 14 to 17):** Galvanized structure mounting → Inverter & earthing (<5 Ω) → 10% On Installation payment → Net-meter grid sync.

**5. Closure & Subsidy Pipeline (Steps 18 to 20):** 10% Final balance collected (Balance = Rs. 0) → PM Surya Ghar claim submission → DBT credit to customer bank.

### THE 3 MANDATORY PAYMENT GATES (80 / 10 / 10 CASHFLOW PROTECTION):

**1. First Payment (80% Advance — Step 07): Paid upon order confirmation. Hard Gate: No factory PO or material dispatch is permitted until Accounts confirms the bank receipt.**

**2. Second Payment (10% On Installation — Step 16):** Collected immediately after mechanical mounting, inverter placement, and DC/AC cabling are finished on site before the utility net-meter inspection.

**3. Final Payment (10% Final Settlement — Step 18):** Cleared upon bi-directional net-meter sync. Customer balance becomes **Rs. 0.00**.

*Note: Central DBT subsidy (Rs. 30,000 to Rs. 78,000) is paid directly by government to customer bank account. The customer pays 100% of the project contract value directly to NIRT.*

# QUALIFYING LEADS & FEASIBILITY AUDIT (STEPS 02 & 03)

## SECTOR VIABILITY CRITERIA, DROP REASONS & 4-POINT PHOTO SURVEY

Sales and engineering evaluate every project based on practical sector requirements. Separating viable clients from unviable proposals keeps field operations efficient and profitable:

### STEP 02: LEAD QUALIFICATION CRITERIA & OUTCOMES:

**1. QUALIFIED PROSPECTS (PROCEED TO STEP 03 SITE SURVEY):**

- **Residential Homes (PM Surya Ghar):** Property owner with active EB connection seeking to zero out power bills.
- **Commercial Complexes:** Showrooms, hospitals, schools, and offices running heavy daytime loads.
- **Industrial Plants & Mills:** Manufacturing sheds and spinning mills seeking to reduce peak HT/LT tariff slabs.
- **Farm Lands & Agro-Solar:** Borewell agricultural pumps (PM-KUSUM) and ground solar installations.

**2. UNQUALIFIED / DROPPED PROPOSALS (LOG REASON IN CRM — DO NOT PROCEED):**

- **Reason 1: Disputed Property / No Landlord NOC: Title in dispute or tenant without registered owner consent.**
- **Reason 2: Structurally Unsafe Roof: Fragile or corroded metal sheets unable to carry structure and wind load.**
- **Reason 3: Permanent Sun Obstruction: Heavy shading from tall adjacent buildings or dense trees.**
- **Reason 4: Local Transformer Saturation: Local TANGEDCO distribution transformer has zero remaining solar quota.**
- **Reason 5: Out of Territory: Outside operational logistics radius.** ***(Note: If entered incorrectly, use Edit with Reason to restore).***

### STEP 03: ON-SITE TECHNICAL FEASIBILITY & MANDATORY 4 PHOTOS:

Before preparing the commercial proposal, the surveyor audits 3 physical engineering parameters and uploads 4 mandatory photos:

**1. Sanctioned Electrical Load:** Record consumer number, sanctioned kW, 1-Phase vs 3-Phase LT / HT, and breaker capacity.

**2. Tape Dimensions & Shadow:** Measure terrace length x width (usable sq.ft), mumty/tank height, and South orientation.

**3. Cable Routing & Inverter Wall:** Measure DC/AC cable run meters and designate a sheltered wall for inverter and DB boxes.

**MANDATORY 4 PHOTOS UPLOADED:** [1] Panoramic Roof Area, [2] TANGEDCO EB Energy Meter, [3] Main Distribution Board (DB), [4] Inverter Mounting Wall.

*Correction Rule: If dimensions or photos are wrongly recorded, submit Edit with Reason in CRM to update.*

# ROLES & DAILY WORK: MANAGEMENT & SALES

**SYNCHRONIZED OPERATIONAL ROLES, OWNED STEPS & STAGE GATES**

## ROLE 1: MANAGING DIRECTOR / BUSINESS OWNER

**Department & Everyday Job: Executive Strategy & Project Governance**
**Operating Pipeline: All 5 Pipelines (Master Oversight) | Owned Pipeline Steps: All 20 Steps (Executive Authority)**

| Rs. 48.2 Lakhs | 42 Sites | 22.4 % | Rs. 0.00 |
|---|---|---|---|
| Monthly Sales Booked | Active Ongoing Sites | Profit Margin Yield | Cash Stuck in Field |

**What This Role Sees on Dashboard (Live Status & Data):**

• **Monthly Revenue & Margin Trend** *(Bar-Line Chart)*:
May: Rs. 32L (21%) | Jun: Rs. 36L (22%) | Jul: Rs. 41L (22%) | Aug: Rs. 44L (23%) | Oct: Rs. 48.2L (22.4%)

• **Sales by Customer Category** *(Donut Chart)*:
Residential PM Surya Ghar: 62% | Commercial LT/HT: 28% | Solar Water Pumps & Farms: 10%

• **5-Pipeline System Balance** *(Stacked Bar)*:
P1 Sales: 18 leads | P2 Approvals: 9 | P3 Procurement: 6 | P4 Execution: 12 | P5 Subsidy: 7

**Operational Execution (Inputs, Uploads & Stage Gates):**

• **Steps 01 to 20 Governance:** Inspect 5-Pipeline Master Board daily; track projects pending Step 07 Advance or Step 17 Net-Meter Sync.

• **Step 05 Discount Approval:** Authorize commercial discounts or custom payment terms for factories/complexes before sales dispatches quote.

• **Audit Log Oversight:** Review immutable system audit ledger daily; inspect all 'Edit with Reason' entries for complete operational transparency.

## ROLE 2: FIELD SALES REPRESENTATIVE

**Department & Everyday Job: Customer Acquisition & Quotation Sizing**
**Operating Pipeline: Pipeline 1: Sales & Feasibility | Owned Pipeline Steps: Steps 01, 02, 05, 06**

| 28 Customers | 8 Sites | 6 Inquiries | Rs. 64.5 Lakhs |
|---|---|---|---|
| New Leads Assigned | Surveys Booked | Unqualified / Dropped | Deals Won (80% Paid) |

**What This Role Sees on Dashboard (Live Status & Data):**

• **Sales Funnel Progression** *(Funnel Chart)*:
Inquiries (28) → Qualified (22) → Survey Done (14) → Quote Sent (10) → Won / 80% Paid (6)

• **Step 02 Disqualification Audit** *(Donut Chart)*:
No Property Title/NOC (35%) | Unsafe Roof (25%) | Grid Saturated (20%) | Extreme Shadow (20%)

• **Lead Source Performance** *(Bar Chart)*:
WhatsApp Campaigns (32%) | Customer References (28%) | Direct Walk-ins (22%) | Electrical Contractors (18%)

**Operational Execution (Inputs, Uploads & Stage Gates):**

• **Step 01 (Lead Inward):** Input customer name, phone, circle, and EB consumer number. Real-time engine prevents duplicate entries.

• **Step 02 (Qualification):** Verify viability. Assign Qualified (Cat A/B/C) or archive as Unqualified (Cat D Dropped) with mandatory reason.

• **Steps 05 & 06 (Quote & Order):** Generate official BOM proposal, secure signed Work Order, allocate Project ID, and route to Accounts for Step 07.

• **Edit Rule:** If customer contact, address, or load was typed wrongly, click 'Edit' and enter a mandatory reason (logged to PMO Audit).

# ROLES & DAILY WORK: SURVEY & PROCUREMENT

**SYNCHRONIZED OPERATIONAL ROLES, OWNED STEPS & STAGE GATES**

## ROLE 3: SITE SURVEY ENGINEER

**Department & Everyday Job: Field Feasibility & Electrical Audit**
**Operating Pipeline: Pipeline 1: Sales & Feasibility | Owned Pipeline Steps: Step 03 (Survey Dossier)**

| 6 Scheduled | 22 Sites | 2 Sites | 2.4 Days |
|---|---|---|---|
| Surveys Today | Surveys Completed MTD | Unfeasible / Rejected | Avg Time per Survey |

**What This Role Sees on Dashboard (Live Status & Data):**

- **Roof & Ground Distribution** *(Donut Chart)*:
  RCC Concrete Flat (55%) | Industrial Metal Shed (25%) | Farm Land / Ground (10%) | Superstructure (10%)
- **Electrical Supply Phase** *(Bar Chart)*:
  Single-Phase LT: 50% | Three-Phase LT: 40% | High Tension (HT 11kV/22kV): 10%
- **Feasibility Approval Ratio** *(Pie Chart)*:
  Feasible Approved: 91% (20 sites) | Load Enhancement Needed: 6% (1 site) | Structurally Rejected: 3% (1 site)

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **Step 03 (Physical Survey):** Record terrace length x width, usable shadow-free area, mumty height, and South azimuth.
- **Step 03 (Mandatory 4 Photos):** Capture and upload high-res field photos of [1] Roof, [2] EB Meter, [3] DB Panel, and [4] Inverter Wall.
- **Step 04 Handoff:** Measure AC/DC cable distances and submit survey dossier for engineering SLD and CAD layout release.
- **Edit Rule:** If obstacle measurements or photos need correction, use 'Edit with Reason' to log modifications in the audit trail.

## ROLE 4: PURCHASE & WAREHOUSE MANAGER

**Department & Everyday Job: Procurement, QC & Inventory Logistics**
**Operating Pipeline: Pipeline 3: Procurement & Store | Owned Pipeline Steps: Steps 11, 12, 13**

| 5 Projects | 180 kW | 14 Units | 100 % Orders |
|---|---|---|---|
| Orders Ready to Pack | Solar Panels in Stock | Inverters in Stock | Advance 80% Verified |

**What This Role Sees on Dashboard (Live Status & Data):**

- **Warehouse Inventory Stock Level** *(Bar Chart)*:
  Mono PERC Panels: 180 kW (Min 50 kW) | Inverters: 14 (Min 5) | DC Cable: 1,200m | Structure: 22 Sets
- **Material Dispatch Turnaround** *(Donut Chart)*:
  Same-Day Dispatch (40%) | Within 24 Hours (50%) | Within 48 Hours (10%)
- **Step 07 Gatekeeper Enforcement** *(Bar Chart)*:
  100% of dispatched BOMs verified with Accounts 80% Advance UTR (Zero unfunded dispatches)

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **Step 11 (Vendor PO):** Issue Purchase Orders to Tier-1 module, inverter, and structure vendors only after Accounts clears Step 07 Advance.
- **Step 12 (Warehouse QC & Inward):** Verify supplier deliveries, log serial numbers, and generate Goods Receipt Note (GRN) tagged to Project ID.
- **Step 13 (Site Dispatch):** Kit full BOM, print stamped Delivery Challan, and dispatch vehicle to customer site.
- **Edit Rule:** If PO specs or serial numbers were mistyped, submit 'Edit with Reason' to record the correction in the audit log.

# ROLES & DAILY WORK: INSTALLATION & TANGEDCO

**SYNCHRONIZED OPERATIONAL ROLES, OWNED STEPS & STAGE GATES**

## ROLE 5: EPC OPERATIONS MANAGER

**Department & Everyday Job: Field Construction & Quality Assurance**
**Operating Pipeline: Pipeline 4: EPC Site Execution | Owned Pipeline Steps: Steps 14 & 15**

| 12 Sites | 8 Sites | 5 Sites | 420 Days |
|---|---|---|---|
| Active Work Sites | Structure & Panels Done | Inverter Tested & Wired | Zero Site Accidents |

**What This Role Sees on Dashboard (Live Status & Data):**

- **Active Site Milestones** *(Stacked Bar)*:
  Structure Erected (12 sites) → Panels Clamped (10 sites) → Inverter Wired (8 sites) → Tested (5 sites)
- **Earthing Resistance Log (≤ 5 Ω)** *(Donut Chart)*:
  Below 2 Ohms: 60% | 2 to 4 Ohms: 35% | 4 to 5 Ohms: 5% | Above 5 Ohms (Fail): 0%
- **Weekly Site Crew Output** *(Bar Chart)*:
  Team Alpha (Urban Sites): 18 kW | Team Beta (Industrial Sheds): 15 kW | Team Gamma (Rural/Farms): 12 kW

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **Step 14 (Civil & Mechanical):** Anchor GI structure legs, torque-clamp solar modules, and waterproof roof penetrations.
- **Step 15 (Electrical & Inverter):** Mount inverter, run DC/AC cables in UV conduits, and test chemical earth pit resistance (< 5 Ω).
- **Step 16 Milestone Alert:** Complete electrical checklist, record string VOC/ISC, and alert Accounts to collect 10% On Installation.
- **Edit Rule:** If torque values or earth resistance readings are entered wrongly, use 'Edit with Reason' to log corrected data.

## ROLE 6: TANGEDCO LIAISON OFFICER

**Department & Everyday Job: Utility Approvals & Grid Synchronization**
**Operating Pipeline: Pipeline 2 (Approvals) & Pipeline 4 (Sync) | Owned Pipeline Steps: Steps 08, 09, 10, 17**

| 16 Sites | 11 Sites | 7 Sites | 12.4 Days |
|---|---|---|---|
| Applications Filed | Feasibility Approved | Meters for Install | Avg Approval Time |

**What This Role Sees on Dashboard (Live Status & Data):**

- **TANGEDCO Utility Funnel** *(Funnel Chart)*:
  Form 1 Submitted (16) → Feasibility Approved (11) → Meter Fee Paid (9) → Meter Sync (7)
- **Approval Speed by Section Office** *(Bar Chart)*:
  Section Office A: 10 Days | Section Office B: 12 Days | Section Office C: 14 Days | Section Office D: 13 Days
- **Bi-Directional Meter Readiness** *(Donut Chart)*:
  Single-Phase Net Meters Ready (12 units) | Three-Phase Net Meters Ready (8 units)

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **Step 08 & 09 (KYC & DISCOM Filing):** Assemble KYC documents in digital locker; submit online Form 1 application to TANGEDCO portal.
- **Step 10 (Feasibility Sanction):** Coordinate with local Assistant Engineer (AE) for transformer load inspection; upload sanction letter.
- **Step 17 (Net Metering Sync):** Witness bi-directional net-meter installation, secure signed Joint Commissioning Report (JCR), and upload to CRM.
- **Edit Rule:** If DISCOM application number or sanction reference was mistyped, submit 'Edit with Reason' to correct.

# ROLES & DAILY WORK: ACCOUNTS & SUBSIDY

**SYNCHRONIZED OPERATIONAL ROLES, OWNED STEPS & STAGE GATES**

## ROLE 7: FINANCE & ACCOUNTS OFFICER

**Department & Everyday Job: Commercial Gatekeeper & Milestone Billing**
**Operating Pipeline: Commercial Gatekeeper (Pipelines 2, 4, 5) | Owned Pipeline Steps: Steps 07, 16, 18 (Hard Gates)**

| Rs. 78.4 Lakhs | 18 Projects | Rs. 0.00 | 100 % Clean |
|---|---|---|---|
| Cash Collected MTD | 80% Advances Cleared | Overdue Customer Balance | Tax GST Reconciled |

**What This Role Sees on Dashboard (Live Status & Data):**

- **80 / 10 / 10 Cash Inflow Ratio** *(Donut Chart)*:
  80% Upfront Advance (78% of cash) | 10% Assembly Balance (12%) | 10% Net-Meter Final (10%)
- **Payment Mode Breakdown** *(Bar Chart)*:
  Bank NEFT / RTGS Transfers: 68% | Digital UPI / QR: 22% | Cheque Clearance: 10%
- **Customer Receivables Aging** *(Bar Chart)*:
  Current / Zero Balance: 96% | Under 7 Days Pending: 4% | Over 7 Days Past-Due: 0%

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **Step 07 (80% Advance Gatekeeper):** Match bank NEFT/RTGS credit, enter UTR, and clear Stage 7 to unlock warehouse material procurement.
- **Step 16 (10% On Installation):** Log customer payment receipt as soon as mounting and inverter cabling are completed on site.
- **Step 18 (10% Final Balance):** Collect final 10% on net-meter sync, reconcile balance to Rs. 0.00, and issue final tax invoice.
- **Edit Rule:** If UTR reference or payment amount is entered with a typo, use 'Edit with Reason' to log correction in audit trail.

## ROLE 8: CENTRAL SUBSIDY COORDINATOR

**Department & Everyday Job: PM Surya Ghar Consumer DBT Administration**
**Operating Pipeline: Pipeline 5: Closure & Subsidy | Owned Pipeline Steps: Steps 19 & 20**

| 34 Claims | 26 Grants | Rs. 19.8 Lakhs | 21.0 Days |
|---|---|---|---|
| National Portal Claims | Subsidy Approved / Paid | Subsidy Benefit Received | Avg Days to Bank Credit |

**What This Role Sees on Dashboard (Live Status & Data):**

- **Subsidy Claims by Tier** *(Donut Chart)*:
  1 kW (Rs. 30,000 benefit): 15% | 2 kW (Rs. 60,000 benefit): 25% | 3 kW+ (Rs. 78,000 benefit): 60%
- **National Portal Stage Speed** *(Funnel Chart)*:
  Registered (34) → Feasibility Cleared (31) → JCR Uploaded (28) → Subsidy Disbursed (26)
- **Aadhaar Bank Link Verification** *(Pie Chart)*:
  Verified Bank Account (97%) | Correction / Re-upload Needed (3%) | Rejected: 0%

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **Step 19 (National Portal Claim):** Register eligible residential customer on PM Surya Ghar portal using electricity consumer number and Aadhaar.
- **Step 19 (Documentation Upload):** Upload signed JCR commissioning report, installed bi-directional meter photo, and inverter serial number.
- **Step 20 (DBT Verification):** Monitor central DBT transfer queue; confirm credit with customer bank statement and log transaction UTR.
- **Edit Rule:** If portal claim reference or bank IFSC was mistyped, submit 'Edit with Reason' to correct.

# PROJECT COORDINATOR & RESPONSIBILITY MATRIX

**CENTRAL STAGE-GATE GOVERNANCE ACROSS STEPS 01 TO 20**

## ROLE 9: PROJECT COORDINATOR (CENTRAL PMO)

**Department & Everyday Job: Central Project Management Office (PMO)**
**Operating Pipeline: Central Stage-Gate Governance | Owned Pipeline Steps: All 20 Steps & Step 20 Triple Closure**

| 96.2 % | 0 Sites | 184 Events | 4.9 / 5.0 |
|---|---|---|---|
| On-Time Completion Rate | Sites Delayed / Stuck | Immutable Audit Logs MTD | Customer Feedback Score |

**What This Role Sees on Dashboard (Live Status & Data):**

- **Projects Across 5 Pipelines** *(Horizontal Bar)*:
  P1 Sales: 18 | P2 Approvals: 9 | P3 Procurement: 6 | P4 Execution: 12 | P5 Subsidy: 7
- **Average Handover Time (TAT)** *(Bar Chart)*:
  Lead to Survey: 1.8 Days | Survey to Quote: 1.0 Day | Advance to Pack: 2.2 Days | Install to Meter: 12.4 Days
- **Departmental On-Time Score** *(Scoreboard)*:
  Sales (98%) | Survey (96%) | Warehouse (99%) | Site EPC (95%) | Liaison (92%) | Accounts (100%)

**Operational Execution (Inputs, Uploads & Stage Gates):**

- **SLA Velocity Oversight:** Track cross-department handoffs (Sales to Survey ≤ 24h, Accounts to Warehouse ≤ 24h, Install to Meter ≤ 12d).
- **Step 20 (Triple Project Closure):** Verify Technical Closure (Step 17), Commercial Closure (Step 18), and Subsidy Closure (Step 20).
- **Audit Trail Oversight:** Inspect daily immutable audit log to review all staff actions, edits with reason, and stage gate clearances.

**DEPARTMENTAL RESPONSIBILITY MATRIX (MAPPED TO STEPS 01 TO 20):**

| Department | Primary Role | Owned Steps (Page 1) | Key Deliverable / Gate Output |
|---|---|---|---|
| 1. Sales | Sales Executive | Steps 01, 02, 05, 06 | Signed Work Order & Project ID Generated |
| 2. Engineering | Site Surveyor & Design Eng | Steps 03, 04 | Survey Dossier (4 Photos) & Stamped SLD |
| 3. Accounts | Accounts Executive | Steps 07, 16, 18 | 80% Advance Clearance & Balance = Rs. 0 |
| 4. Liaison | TANGEDCO Officer | Steps 08, 09, 10, 17 | DISCOM Sanction & Net-Meter Grid Sync |
| 5. Purchase & Stores | Warehouse Manager | Steps 11, 12, 13 | Factory PO, Inward GRN & Delivery Challan |
| 6. Installation (EPC) | EPC Site Operations | Steps 14, 15 | Structure Erected & Inverter Earthing < 5 Ω |
| 7. Subsidy Desk | Subsidy Coordinator | Steps 19, 20 | National Portal Claim & DBT Bank Credit |
| 8. Central PMO | Project Coordinator | Steps 01 to 20 Oversight | Triple Project Closure & Audit Integrity |

# SITE VERIFICATION, CLOSURES & AUDIT LOG

## THE 3 MANDATORY CLOSURES & FORENSIC AUDIT TRAIL ARCHITECTURE

### CORE FIELD VERIFICATION GUIDELINES (TIED TO PIPELINE GATES):

- **Before Quotation (Step 03):** Surveyor inspects EB meter sanctioned load, roof shadow azimuth, and inverter wall location.
- **Before Procurement (Step 07):** Accounts confirms bank UTR credit for the 80% Advance payment.
- **During Assembly (Steps 14 & 15):** EPC team checks structure bolt torque, UV conduit protection, and earthing resistance (≤ 5 Ω).
- **During Grid Sync (Step 17):** Liaison officer witnesses TANGEDCO bi-directional net-meter installation and gets official inspection report signed.

### THE 3 MANDATORY PROJECT CLOSURES (STEP 20 FORMAL COMPLETION):

**1. Technical Closure (Step 17):** Solar panels generating power, net-meter synchronized, and TANGEDCO AE signs Joint Commissioning Report.
**2. Commercial Closure (Step 18):** Customer has cleared 100% contract value (80% + 10% + 10%). Accounts confirms customer balance is **Rs. 0.00**.
**3. Subsidy Closure (Step 20):** Commissioning report uploaded to National Portal, and Central DBT subsidy is credited into customer's bank account.

### WHAT IS HANDED OVER TO THE CUSTOMER IN THEIR HARD-BOUND DOSSIER:

- **25-Year Solar Panel Warranty:** Factory serial numbers matched with installed TopCon / Mono-PERC panels.
- **Inverter 5/10-Year Warranty Card:** OEM warranty certificate with customer Wi-Fi monitoring portal credentials.
- **Single Line Wiring Diagram (SLD):** Stamped electrical schematic showing DC/AC conduits and safety disconnects.
- **TANGEDCO Net-Meter Inspection Report:** Official utility bi-directional commissioning record.
- **Emergency Solar Manual:** How to clean panels safely with water, and how to operate the DC isolation safety switch.

### SYSTEM AUDIT LOG & COMPLIANCE GOVERNANCE (PMO VISIBLE ONLY — NO DELETE OPTION):

**1. Strictly Append-Only (Zero Delete Option):** Every operational action across all 9 roles is written permanently to the system audit ledger. No role, technician, or administrator has permission to delete historical audit records.
**2. Visible Only to PMO & Executive Leadership:** The audit trail is restricted strictly to the Project Coordinator (Central PMO) and Managing Director. It is completely hidden from operational roles (Sales, Survey, Warehouse, EPC Ops, Liaison, and Accounts).
**3. 'Edit with Reason' Tracking:** If a user enters wrong data at any step (e.g. wrong EB number, incorrect load, or typo in UTR), they can edit by submitting a mandatory reason. The system forensically records Date/Timestamp (IST), Staff Name, Role, Project ID, Previous Value → New Value, and Reason for Edit.